

CARNEGIE MELLON UNIVERSITY • SCHOOL OF COMPUTER SCIENCE

Executive Education: Agentic AI Program • Capstone Project

Checkpoint 1.1: Problem Framing & Initial Agent Design

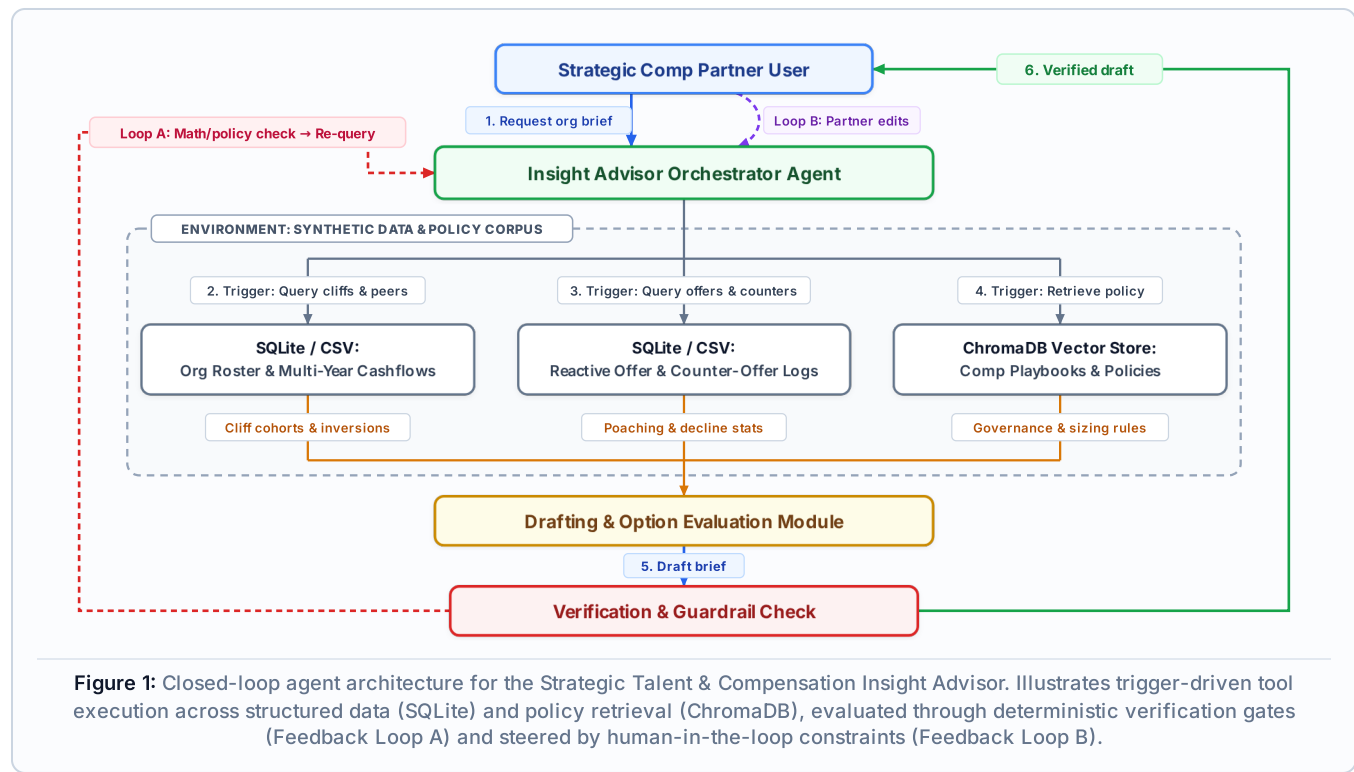
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Project: Strategic Comp Insight Advisor

Track: RESEARCH ASSISTANT

Submission: MODULE 1 • CHECKPOINT 1.1

SYSTEM ARCHITECTURE DIAGRAM



WRITTEN SUBMISSION

1. The agent, the problem, and the intended user

The **Strategic Talent and Compensation Insight Advisor** is a Research Assistant agent built for an enterprise **Strategic Compensation Partner** who advises Engineering Vice Presidents and Lead People Partners. Today, compensation teams often operate in two disconnected silos. A transactional Offers and Counter-Offer team handles external hiring and reactive retention when employees receive competing offers, while Strategic Compensation Partners manage proactive retention budgets, annual equity cycles, and organizational health.

Before a monthly talent review with a VP, the Compensation Partner must manually stitch together spreadsheets of internal employee compensation, multi-year equity vesting schedules, recent offer decline logs, and evolving policy documents. Because this manual synthesis takes hours of spreadsheet work, partners struggle to spot early links between external hiring friction and internal retention risk, and executives receive dense tables rather than clear, actionable insights. The agent solves this by investigating structured workforce data alongside policy documentation to produce a concise, verified executive decision brief.

WRITTEN SUBMISSION (CONTINUED)

2. Why a standalone LLM or simple prompting is insufficient

A standalone LLM fails at this task for four fundamental architectural reasons:

Context window degradation and data scale

Dumping thousands of employee roster rows, multi-year vesting schedules, and dozens of policy documents into a single prompt exceeds smaller model context windows and degrades reasoning accuracy in larger models.

Exact arithmetic vs. probabilistic text

Compensation analysis requires exact aggregations, percentile rankings, year-over-year cashflow drop calculations, and budget caps that LLMs hallucinate without deterministic data tools.

Dynamic, multi-step conditional investigation

Investigating a talent hotspot requires iterative hypothesis testing rather than a single prompt or a rigid, fixed pipeline. The agent must dynamically choose which dataset or tool to query next based on intermediate results: detecting where internal equity cliffs exist, checking whether that same job family faces declining offer acceptance rates or rising counter-offers, and retrieving the governance policy that determines whether to intervene through offer bands or proactive retention grants.

Verification before executive delivery

Outputs shared with VPs require automated validation against source numbers and policy constraints before a human partner reviews them.

3. The environment: documents, data sources, tools, and users

The agent operates within a local Python environment using a Gradio interface and a two-tier model router supporting OpenRouter cloud models and local Ollama models. Offloading calculations and retrieval to deterministic tools allows smaller 8B to 12B models to handle structured tool calls reliably, while multi-step action selection and trade-off synthesis route to a stronger reasoning model. It interacts with strictly synthetic data representing a 2,000-person technology organization:

Structured data sources

Three relational tables stored in SQLite and CSV format: an *Org Roster & Cashflow Table* containing synthetic salaries, peer percentiles, performance ratings, and four-year vesting schedules; a *Reactive Offers & Counters Log* tracking candidate offer acceptances, declines, competing employers, and counter-offer outcomes; and a *Department Budget Ledger*.

Unstructured knowledge base

A ChromaDB vector store indexing synthetic compensation policy PDFs, guidelines defining the boundary between reactive counter-offers and proactive retention, and historical executive briefing templates.

Tools and users

Deterministic Python/Pandas query tools that return compact Markdown table summaries to preserve context window efficiency, a semantic search tool over ChromaDB, a report generator, and the Compensation Partner user who steers the investigation.

WRITTEN SUBMISSION (CONTINUED)

4. The actions the agent needs to take and their triggers

Rather than executing a fixed pipeline, the agent dynamically selects its next action based on intermediate state and trigger conditions:

- 1 Scan and aggregate org health metrics.** **TRIGGERED** when the user selects a VP organization and requests a talent review. The agent calls the roster analysis tool to identify cohorts with projected year-over-year cashflow drops exceeding 15 percent or peer positioning below the 25th percentile.
- 2 Cross-examine reactive market signals.** **TRIGGERED** dynamically when Action 1 flags an at-risk job family or location. The agent queries the Offers and Counters Log to measure whether offer acceptance rates have dropped or counter-offer volume has spiked in that same segment.
- 3 Retrieve governance and handoff rules.** **TRIGGERED** when intermediate results confirm both reactive market pressure and internal retention risk in the same cohort. The agent queries ChromaDB for policy rules governing whether the issue should be addressed by the Offers team adjusting hiring bands or the Client Partner deploying proactive retention equity.
- 4 Draft the executive insight brief.** **TRIGGERED** once data aggregation and policy retrieval complete. The agent synthesizes a one-page brief summarizing the risk, comparing intervention options against available budget, and citing exact source metrics.

5. How feedback guides behavior across steps

The system relies on three feedback loops to guide action selection across steps:

Tool execution and intermediate state feedback

If a SQLite or Pandas query returns an empty cohort, a syntax error, or a sample size too small for statistical relevance, the feedback signal prompts the agent to broaden its filter criteria, such as expanding from a single sub-team to the broader job family, or skip redundant downstream queries before proceeding.

Self-verification and policy audit loop

Before displaying the brief, a verification step compares every dollar figure and headcount number in the drafted text against the raw tool outputs and checks that proposed retention spend does not exceed the remaining budget in the ledger. Any discrepancy triggers a targeted regeneration of the flawed section.

Human-in-the-loop partner refinement

When the Compensation Partner reviews the brief in the UI and adjusts a constraint, such as lowering the budget cap or excluding employees promoted within the last six months, the agent captures that feedback, updates its session memory, re-runs the affected tool calculations, and revises the trade-off recommendations.